



GenAI in Investment Research

Introduction for analysts

Introduction to GenAI and LLMs

Understanding the technologies reshaping investment research



What is Generative AI?

AI that creates new content like text, images, and music. Think of it like a creative assistant that can write stories or compose melodies.



Large Language Models (LLMs)

Advanced AI systems trained on vast amounts of text data to understand and generate human-like language. They're like highly sophisticated text prediction engines.



How They Work

Using deep learning techniques, these models analyze patterns in data to predict and generate content, similar to how humans learn from examples and create new ideas.



Real-World Applications

From chatbots and content creation to advanced data analysis in investment research, these technologies are transforming how we work with information.

GenAI in Finance and Investment Research

Current Landscape of GenAI in Finance

Growing Adoption

GenAI is increasingly being adopted in investment research for tasks like information retrieval, idea generation, data analysis and report generation.

Competitive Landscape

Major financial institutions, large data providers and AI-native startups are building in the space and competing for analysts' ever fragmented attention.

Lack of Expertise

There is interest to build internally, but few financial institutions have the resources and expertise. They are looking to partner with external providers.

Impact on Investment Practices



Enhanced Efficiency

Analysts can automate repetitive tasks, enabling stronger focus on strategic decision-making



Data-Driven Insights

Extract structured insights from vast unstructured datasets for deeper market understanding



Single Source of Truth

A centralized platform that consolidates multiple data sources and information streams to create a unified research hub for analysts

! The Analyst Today



Siloed information sources



Manual information gathering



Report writing from scratch



Manual data cleaning & analysis



of analyst time spent on manual tasks that can be
augmented by AI

The AI-Augmented Analyst



Focus on deep research and idea generation

Shift from routine tasks to generating unique insights and deeper analysis



Faster research time to market

Near-instant analysis and drafting of market events and company developments



Expanded research coverage

Dramatically increased breadth and depth of company and sector coverage



Higher client engagement

More time for high-value client interactions and customized research delivery



of time spent on high-value research, idea generation and client interactions

The Challenge To Effective Adoption of GenAI



Data Privacy Concerns

Ensuring confidential data remains secure when using LLMs while maintaining compliance and trust.



Regulatory Compliance

Complex regulations around research and distribution pose challenges for LLM adoption in investment workflows.



Domain-Specific Adjustment

Adapting LLMs to understand financial terminology and industry-specific contexts requires significant resources.



Reliability and Accuracy

Balancing high accuracy requirements with inherent limitations of AI models in investment research.



Trust in AI

Building trust in AI-generated insights when LLMs lack clear explainability in their decision-making process.



Learning Curve for Analysts

Mastering prompt engineering and identifying optimal use cases requires significant training and adaptation.

Recent Research and Developments



Expanded Context Windows

Larger context windows enable models to handle bigger documents and more data, allowing for easier and more reliable synthesis of information.



Advancements in AI Reasoning

OpenAI's O1 model demonstrates advanced reasoning, with research also being done in the area of integrating LLMs with Symbolic AI to enhance interpretability.



Finance-specific Benchmarks

FinanceBench is the industry benchmark for financial Q&A LLM accuracy. Performance of LLM applications keep improving, with Quantly achieving a top accuracy of 94%.



Text to Code

Turning text to code techniques improve the reliability of LLM answers, especially when it comes to queries that require calculations of financial metrics or more complicated data analysis.



**GenAI is not a
solution to
everything**



**As with all AI,
rubbish in means
rubbish out**



Fundamentals of Effective Prompting

Definition of a Prompt in the Context of GenAI

What is a Prompt?

- A prompt is the initial input or question given to a GenAI system, serving as the starting point for AI-generated content or analysis.

Prompt Structures

- Prompts can vary from a single line to complex multi-prompt instructions, building on each other for more detailed and nuanced outputs.

Use in Investment Research

- In investment research, prompts range from simple information retrieval queries to complex instructions of multi-step manual workflows, guiding the AI through financial analysis tasks.

The relationship between prompt and output quality

Well-crafted prompts lead to accurate and relevant responses by providing clear guidance to AI systems, directly impacting output quality.

In investment research, precise prompts are crucial for obtaining deep, actionable insights rather than surface-level information.

Principles of Effective Prompting



Be Specific and Concise

Clear, precise prompts yield better results than vague or wordy ones



Use Context-Setting Statements

Provide relevant background information to frame your query appropriately



Break Complex Queries

Split complicated questions into smaller, more manageable parts



Utilize Appropriate Keywords

Include specific terms and industry language relevant to your query



Poor Prompt Example

"Sales of Boeing?"

A vague prompt that leaves a lot to the model's interpretation = Poor result



Good Prompt Example

"You are an equity analyst covering Boeing. Your task is to output a table of Boeing's sales over the past 4 quarters and the YoY% growth rate. Go through the last 4 quarterly reports to find the information."

A well-structured prompt with explicit instructions, a clear context, and a defined desired outcome = Good result

Prompt Structuring: Role-Task-Format Framework



Role

Specify the AI's assumed expertise

Act as a financial analyst



Task

Define the specific analysis or output required

evaluate Company X's latest earnings report



Format

Indicate the desired form of the response

Present your findings in a bullet-point summary

Example:

"As a financial analyst, evaluate Company X's latest earnings report. Present your findings in a bullet-point summary."

Leading Questions Technique

Leading and targeted questions guide the AI towards specific areas of analysis, types of insights and sources of information.

They can help focus the AI's response on particular aspects of financial data, source or company performance.

Interactive Example

Click elements to analyze the structure:

"What are the **three most significant factors** impacting Company Y's **profit margins**, based on their **last four quarterly reports**?"

Specific quantity constraint: three most significant

Analysis focus: factors impacting

Target metric: profit margins

Time period constraint: last four quarterly reports

Iterative Prompting

Use a series of prompts to progressively refine and deepen the analysis. Start with broader queries and then ask follow-up questions based on initial responses.

This approach allows for more thorough and nuanced insights.

Summarize Company A's competitive position

Company A holds a strong market position with 25% market share, industry-leading profit margins, and established distribution networks across key regions.

Follow-up questions:

- What potential threats to this position are evident from recent industry trends?
- How does their market share compare to the top 3 competitors?
- Which regions show the strongest growth potential?

What potential threats to this position are evident from recent industry trends?

Prompt Chaining

Link multiple prompts in a logical sequence to break down complex financial analyses into manageable steps.

Each prompt builds on the previous one, creating a comprehensive analytical process.

Summarize key financial metrics for Company B.

Based on these metrics, identify potential areas of concern.

Key concerns include declining cash flow despite revenue growth, suggesting potential working capital issues. The rising leverage ratio could limit financial flexibility and increase interest burden.

Analyze how these concerns might impact the company's 5-year growth projections.

Common Pitfalls and How To Avoid Them

! **Overly broad or vague prompts**

Generic queries like "Analyze Company X" yield unfocused results, missing critical aspects of financial analysis.

! **Assuming too much AI knowledge**

While AI has broad knowledge, it may lack nuance from knowledge it doesn't have access to. Don't assume it knows everything!

! **Ignoring the AI's limitations**

AI may struggle with real-time data or analysis that requires complex reasoning and assumptions. It's not magic!

! **Failing to provide necessary context**

Without proper background on a company's history or industry trends, AI may miss crucial factors in its analysis.

! **Overreliance on AI-generated information**

Treating AI outputs as definitive without human verification can lead to overlooking critical nuances or potential errors in financial analysis.

! **Expecting it to make decisions for you**

LLMs are great at retrieving, synthesising and summarising information. They should not and will not make investment decisions on your behalf. This is still your job as an analyst!



Advanced Prompting Strategies

for Qualitative Investment Research

Hypothesis-Driven Prompts

Test your hypothesis on historical data:

- Start with an investment hypothesis and design prompts to test it.
- This approach focuses the AI's analysis on specific aspects of your theory.

Interactive Example

Click elements to analyze the structure:

"Analyze Company X's last three years of financial reports to find evidence supporting or refuting the hypothesis that their shift to subscription-based services is improving profit margins."

Historical timeframe: last three years

Data source: financial reports

Core hypothesis: shift to subscription-based services is improving profit margins

Analysis objective: find evidence supporting or refuting

Incorporating Temporal Analysis in Prompts

It allows us to compare information across time:

- Design prompts that explicitly compare information across different time periods to identify trends, shifts, or cyclical patterns in financial data.

Interactive Example

Click elements to analyze the structure:

"Compare the 'Risk Factors' sections of Company Z's annual reports from 2021, 2022, and 2023. Identify any new risks that have been added, risks that have been removed, and changes in the emphasis or description of persistent risks. What might these changes suggest about the evolution of the company's business environment?"

Time period for analysis: annual reports from 2021, 2022, and 2023

Specific document section: Risk Factors sections

Analysis objectives: Identify any new risks that have been added, risks that have been removed, and changes in the emphasis or description of persistent risks

Insight generation: What might these changes suggest about the evolution of the company's business environment

Structuring Comparative Prompts for Cross-entity Insights

Compare information across companies:

- Craft prompts that analyze multiple entities (companies, sectors, etc.) simultaneously to gain comparative insights and identify relative strengths or weaknesses.

Interactive Example

Click elements to analyze the structure:

"Analyze the latest annual reports of the top three companies in the electric vehicle sector. Compare their R&D spending, production scalability, and market expansion strategies."

Analysis scope: latest annual reports of the top three companies in the electric vehicle sector

Comparison metric 1: R&D spending

Comparison metric 2: production scalability

Comparison metric 3: market expansion strategies

Crafting Prompts to Detect Anomalies and Outliers

Pick up inconsistencies across time:

- Structure prompts to identify outliers or unusual patterns in financial data that might signal important changes, risks, or opportunities.

Interactive Example

Click elements to analyze the structure:

"Review Company A's financial statements and earnings call transcripts from the past two years. Identify any unusual fluctuations in key metrics or significant deviations from industry norms, and provide context for each anomaly detected."

Analysis period: past two years

Data sources: financial statements and earnings call transcripts

Anomaly detection focus: unusual fluctuations in key metrics or significant deviations from industry norms

Contextual analysis: provide context for each anomaly detected

Tasks Best Suited for AI Assistance



Large-scale Text Processing

Analyze thousands of financial documents, earnings calls, and reports simultaneously



Pattern Recognition

Identify trends and correlations across multiple companies and time periods



Information Synthesis

Combine insights from diverse sources to create comprehensive market views



Initial Screening

Quickly evaluate potential investment opportunities against defined criteria



Idea Generation

Discover new investment angles by analyzing cross-market relationships



Report Formatting and Creation

Automate the structuring and formatting of research reports and presentations

Combining AI-driven Insights with Human Intuition and Experience



AI as a Starting Point

Using AI insights as an initial guide to diving into your research, setting a foundation for further exploration and analysis



Contextualizing AI Findings

Integrate your expert analyst knowledge and personal insights to interpret AI-generated results, to boost your overall research capacity and quality



Getting Creative

AI allows you to experiment and ask questions that you wouldn't usually ask. Why? It costs a few seconds to get to the answer rather than hours/days



Identifying Alignments and Discrepancies

Spot where AI findings align with or diverge from human expectations, leading to deeper investigation



Broadening Perspectives with AI

Leverage AI to gain new viewpoints, enhancing the scope of traditional analysis and identifying overlooked aspects



Applying Critical Thinking

Employ critical thinking to assess AI recommendations, refining and validating results to ensure accuracy and reliability

**AI enhances, does not replace
analyst judgement**

Integrating AI Into Your Research Process



Start With Low-Stakes Tasks

Test Quantly for data gathering and summarization



Run Parallel Analysis

Compare efficiency and accuracy with traditional methods



Expand Gradually

Progress from simple to complex analyses as confidence grows



Foster Collaboration

Use AI insights as a basis for team discussions



Customize Outputs

Align AI outputs with firm's report formats and requirements



Enhance Efficiency

Leverage AI to uncover new insights and stay competitive

Adapting the Investment Research Role for an AI-Augmented Future



Skill Evolution

- Develop AI prompt engineering expertise
- Enhance data interpretation and validation skills
- Focus on strategic thinking, scenario analysis and asking the right questions



Changing Responsibilities

- Shift from data gathering to insight synthesis
- Increase focus on qualitative factors and unique company dynamics
- Greater emphasis on risk assessment and model stress-testing



New Areas of Expertise

- AI model understanding and limitations
- Integration of alternative data sources
- Ethical considerations in AI-assisted research



Career Impact

- Transition to higher-value analytical and research tasks
- Increased collaboration with data scientists and AI specialists
- Potential for broader coverage and deeper analysis



Quantly

Thank You

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